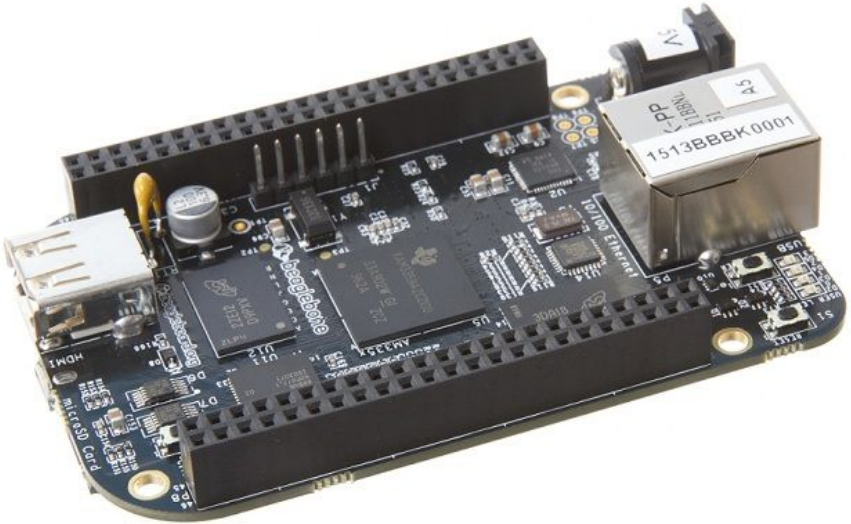
Getting started with Beagleboard

MacOS operating system

Open your Beagle box



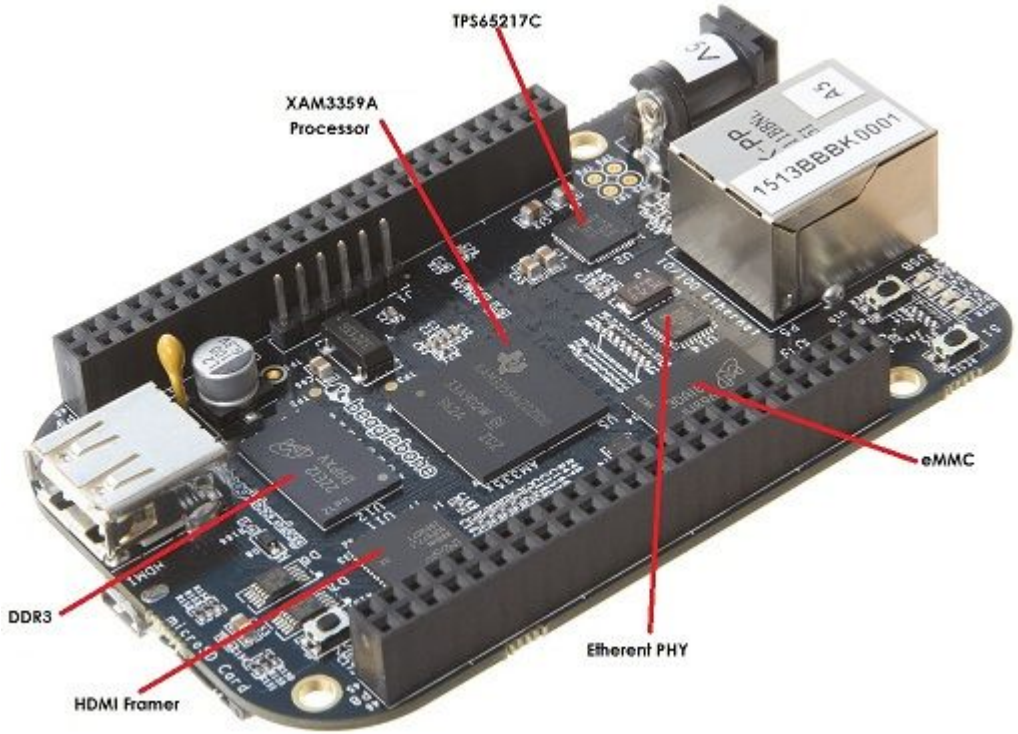
Beaglebone Black board



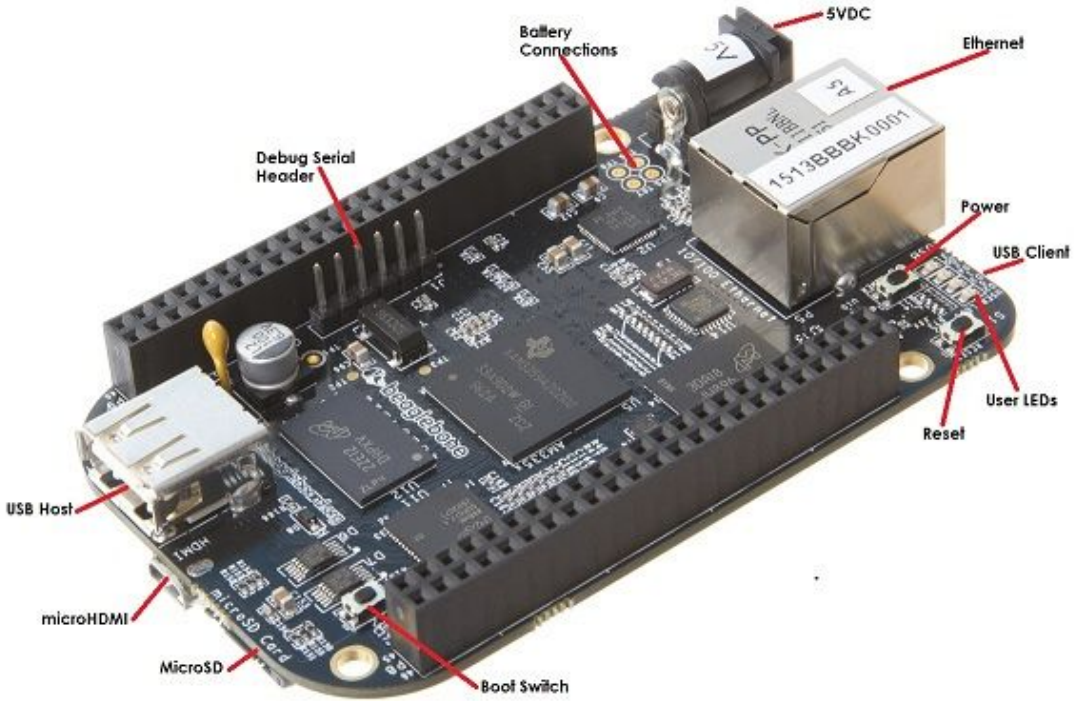
USB mini cable

Open the Beagle box and you will see a Beaglebone Black board and a USB mini cable.

USB mini cable provide a convenient way to provide both power to your Beagle and connectivity to your PC.



BeagleBone Black Key Component Locations



BeagleBone Black Connector and Switch Locations

Update USB CDC Ethernet gadget to use NCM

The USB CDC Ethernet gadget of Beagle is currently configured to use the ECM protocol. Unfortunately this protocol is no longer supported by MacOS hosts. However, the newer NCM protocol is supported.

So if you are using MacOS operating system, you need to follow the Step 1 to Step 5 to update the USB CDC Ethernet gadget configuration scripts to use the NCM protocol instead of the ECM protocol.

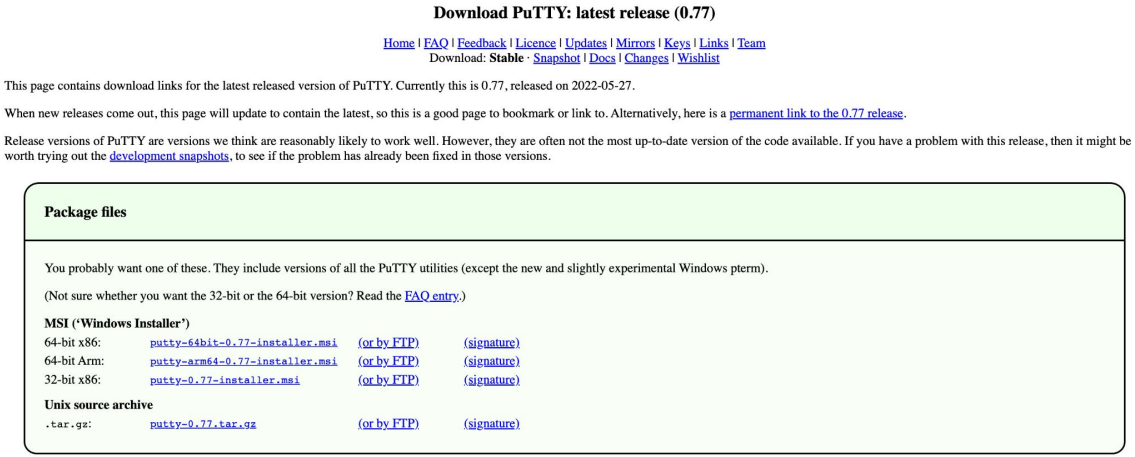
Note: You need to do Step 1 to Step 5 on a windows operating system.

1 Download Putty

Step 1: Go to https://www.putty.org/

Step 2: Click Download PuTTY

Step 3: Choose the package file that matches your operating system and click to download



Download PuTTY: latest release (0.77)

Home | FAQ | Feedback | Licence | Updates | Mirrors | Keys | Links | Team
Download: **Stable** · Snapshot | Docs | Changes | Wishlist

This page contains download links for the latest released version of PuTTY. Currently this is 0.77, released on 2022-05-27.

When new releases come out, this page will update to contain the latest, so this is a good page to bookmark or link to. Alternatively, here is a permanent link to the 0.77 release.

Release versions of PuTTY are versions we think are reasonably likely to work well. However, they are often not the most up-to-date version of the code available. If you have a problem with this release, then it might be worth trying out the development snapshots, to see if the problem has already been fixed in those versions.

Package files

You probably want one of these. They include versions of all the PuTTY utilities (except the new and slightly experimental Windows pterm).

(Not sure whether you want the 32-bit or the 64-bit version? Read the FAQ entry.)

MSI ('Windows Installer')

64-bit x86:	putty-64bit-0.77-installer.msi	(or by FTP)	(signature)
64-bit Arm:	putty-arm64-0.77-installer.msi	(or by FTP)	(signature)
32-bit x86:	putty-0.77-installer.msi	(or by FTP)	(signature)

Unix source archive

.tar.gz:	putty-0.77.tar.gz	(or by FTP)	(signature)

2 Download USB Driver

Step 1: Go to https://beagleboard.org/getting-started

Step 2: Choose the USB Driver file that matches your operating system and click to download

Step 3: Install the USB Driver

Operating System	USB Drivers	Comments
Windows (64-bit)	64-bit installer	If in doubt, try the 64-bit installer first. • **Note #1:** Windows Driver Certification warning may pop up two or three times. Click "Ignore", "Install" or "Run" • **Note #2:** To check if you're running 32 or 64-bit Windows see this: support.microsoft.com/kb/827218. • **Note #3:** On systems without the latest service release, you may get an error (0xc000007b). In that case, please install the following and retry: www.microsoft.com/en-us/download/confirmation.aspx?id=13523. • **Note #4:** You may need to reboot Windows. • **Note #5:** These drivers have been tested to work up to Windows 10
Windows (32-bit)	32-bit installer	
Mac OS X	Network Serial	Install both sets of drivers.
Linux	mkudevrule.sh	Driver installation isn't required, but you might find a few udev rules helpful.

If the installation fails, follow this tutorial to reinstall:
https://support.viewsonic.com/en/support/solutions/articles/33000252416-how-to-install-unsigned-drivers-in-windows-10

3 Power and boot your Beagle

Step 1: Connect the USB mini cable to your BeagleBone Black

Step 2: Connect the other end of the USB cable to a Windows PC

After step 2, you'll see the power (PWR or ON) LED lit steadily. Within a minute or so, you should see the other LEDs blinking in their default configurations. Below is the information for LEDs:

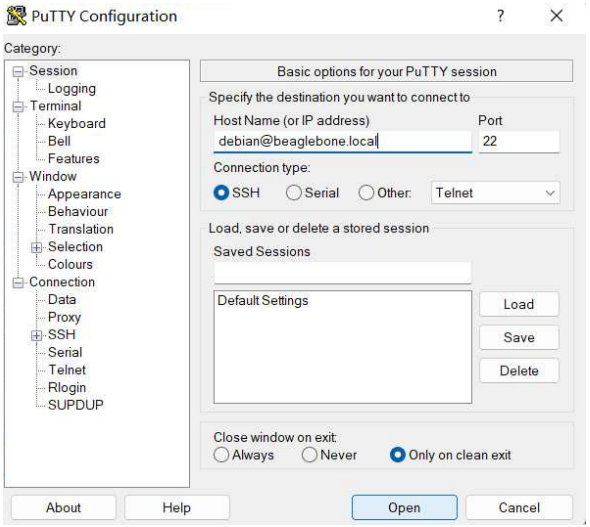
- USR0 is typically configured at boot to blink in a heartbeat pattern
- USR1 is typically configured at boot to light during SD (microSD) card accesses
- USR2 is typically configured at boot to light during CPU activity
- USR3 is typically configured at boot to light during eMMC accesses
- USR4/WIFI is typically configured at boot to light with WiFi (client) network association (*BeagleBone Blue and BeagleBone AI only*)

4 SSH your Beagle

Step 1: Start Putty

Step 2: In the Host Name (or IP address) text box, type **debian@beaglebone.local**

Step 3: In the Port textbox, type **22**

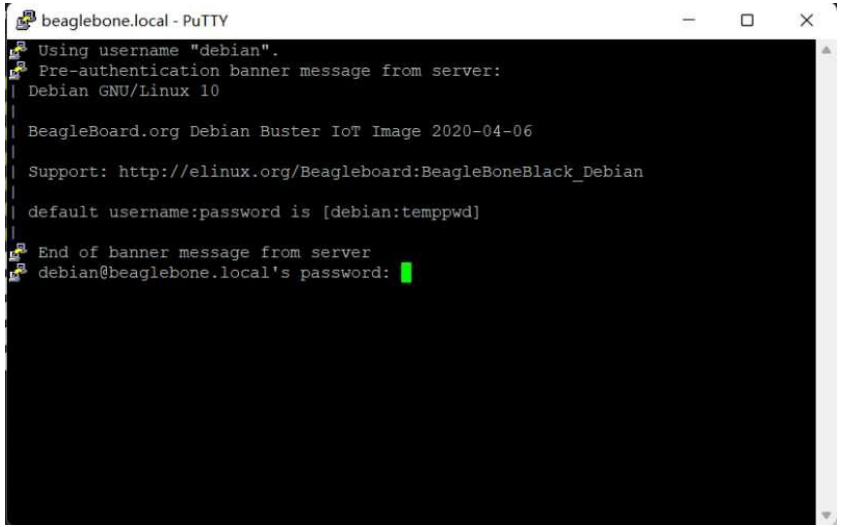


4 SSH your Beagle

Step 4: Click **Open**

Step 5: A PuTTY security alert about the server's host key appears the first time you connect, click **Yes**

Step 6: Type initial password **temppwd** when prompted, and then press **Enter**



5 Update USB CDC Ethernet gadget to use NCM

Follow this tutorial to update USB CDC Ethernet gadget:
https://github.com/RobertCNelson/boot-scripts/commit/b61125c1485bee929340cacc06c85c6fcfd678bc

Step 1: Enter the following command in the console of the putty window:

$ **cd ../../boot**
$ **vim am335x_evm.sh**

Step 2: Press **i** to switch to Insert mode

Step 3: Press ⬇ to go line 484, change “ecm” in lines 484, 485, 486, 488 to “ncm”

Step 4: Press **Esc** to leave Insert mode

Step 5: Type **:wq** and press **Enter** to save file

Step 6: Now you have updated USB gadget in “am335x_evm.sh”, then update “bbai.sh”, “bbai_usb_gadget.sh”, “boot/beagle_x15.sh” in the same way.

6 Unplug your Beagle from Windows pc

Now you have update USB CDC Ethernet gadget to use NCM.

You can eject and unplug the Beagle from Windows and connect the Beagle to MacOS.

7 Connect your Beagle with MacOS

Step 1: Connect the USB mini cable to your BeagleBone Black

Step 2: Connect the other end of the USB cable to your PC

Step 3: Open the Terminal

Step 4: Enter the following command in the Terminal console to ssh your Beagle:
$ **ssh debian@beaglebone.local**

Step 5: Type initial password **temppwd** when prompted, and then press **Enter**

Step 6: Enter the following command in the Terminal console to share network with your Beagle:
$ **sudo dhclient usb1**

Step 7: Type initial password **temppwd** when prompted, and then press **Enter**

Now you have connected your Beagle with your pc.

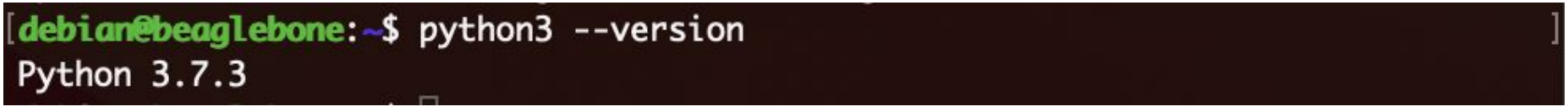
8 Install Python

Step 1: Enter the following command in the Terminal console:

$ **sudo apt update**
$ **sudo apt upgrade**
$ **sudo apt install python3**

Step 2: Enter the following command in the Terminal console to verify the installation:

$ **python3 --version**
You should get the version like **Python 3.7.3** if Python 3 is installed successfully



9 Install Python-OpenCV

Step 1: Enter the following command in the Terminal console:

$ **sudo apt install python3-opencv**

Step 2: Enter the following command in the Terminal console to verify the installation:

$ **python3 -c "import cv2; print(cv2.__version__)"**

You should get the version like **3.2.0** if Python-Opencv is installed successfully

